BIG DATA ANALYTICS FOR PREDICTING CUSTOMER CHURN AND CREDIT BEHAVIOR IN THE FINANCIAL SECTOR

Name: Mahi Chudela
Dept.: Computer Science
School: *University Of Wolverhampton*
Location: Wolverhampton, United Kingdom
Email: M.chudela@wlv.ac.uk

Name: Damilare Ajiboye
Dept: *Computer Science*
School: *University Of Wolverhampton*
Location: Wolverhampton, United Kingdom
Email: d.o.ajiboye@wlv.ac.uk

Abstract

Big Data Analytics helps financial institutions make smarter decisions by analyzing large sets of customer data. Many banks struggle to understand which customers are likely to leave (churn) or how to estimate someone's creditworthiness, which affects loans and services. This project addresses that by building predictive models for churn and credit score estimation using machine learning. The goal is to test and compare several models to find the most accurate and reliable ones for real-world use. We developed and evaluated four classification and four regression models using accuracy, error rates, and feature importance methods. We also used clear visuals and explainable metrics to help understand model decisions. This work supports the financial industry and research community by offering practical, scalable AI solutions for better service and inclusion.

I. BACKGROUND OF THE STUDY

1.1 Generic Information

The advent of digital banking and financial services has led to a massive influx of customer data. Leveraging this data through Big Data Analytics allows financial institutions to uncover behavioral patterns and predict outcomes such as customer churn or credit risk. However, businesses often struggle with choosing the right models and features for their predictive systems.

1.2 Problem Statement

This project focused on using Big Data Analytics to help banks and financial institutions better understand their customers by predicting who might leave (customer churn) and estimating their credit scores. Using real-world banking data, we tested different machine learning models. For classifying churn risk, Random Forest was the most accurate, reaching an 86.36% accuracy rate, while for predicting credit scores, both Linear Regression and Random Forest showed low error margins. We also analyzed which factors (like age or salary) mattered most, and found that even simple data could be powerful. By visualizing results through clear charts, like bar graphs for feature importance *(figure 2)* and choropleth maps to show which countries had the highest churn *(figure 4),* we made it easier to understand the findings. These methods can help financial companies make smarter, faster decisions, improve customer service, and plan where to focus their attention across different regions.

1.3 Aim and Objective

This study aims to leverage machine learning techniques to accurately predict customer churn and estimate credit scores using essential demographic and financial attributes. The primary objective is to evaluate and compare the performance of various classification and regression models, such as Random Forest and Linear Regression, in order to determine the most effective and reliable approaches for practical deployment in real-world financial environments.

1.4 Contributions of the Work

- Developed and compared four classification and four regression models for predicting customer churn and credit scores
- Employed evaluation metrics such as Accuracy, Mean Absolute Error (MAE), Mean Squared Error (MSE), and feature importance to assess performance
- Incorporated K-Fold Cross-Validation to enhance model reliability and generalization
- Used diverse and intuitive data visualization techniques (e.g., bar charts, line graphs, confusion matrices, choropleth maps) to communicate insights clearly
- Demonstrated that the codebase is adaptable to other financial datasets with minimal changes, ensuring scalability
- Aligned predictive insights with practical business strategies, supporting AI deployment in digital banking platforms

1.5 Organization of the Report

This report begins with a review of related literature, followed by a comprehensive methodology section outlining data preparation and model development. Results and discussion analyze the performance of models. The report concludes with insights and potential areas for future work.

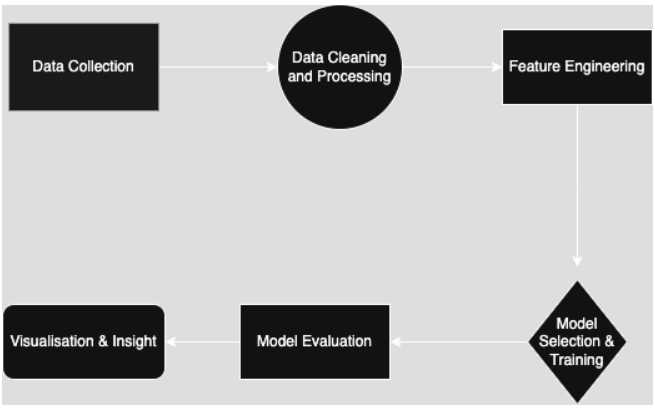
2. Related Work

Prior research has extensively applied machine learning to customer churn prediction, using models such as Logistic Regression, Decision Trees, and Neural Networks. Zhang *et al.* [P] demonstrated that ensemble techniques, particularly Random Forest and Gradient Boosting, consistently outperform individual classifiers in financial churn analysis. Similarly, Shukla and Dey [Q] emphasized the value of explainable models, employing SHAP values to improve transparency in credit score prediction. These efforts highlight the importance of both predictive performance and interpretability in financial analytics.

However, most existing studies tend to isolate classification and regression tasks, lacking comprehensive comparative experimentation on unified datasets. Our work fills this gap by integrating churn classification and credit score regression on the same dataset. Additionally, we align model outcomes with business strategies and visualize insights through interpretable plots, making the results accessible for executive decision-making in financial institutions.

3. Methodology

Our methodology is structured into six distinct phases, as illustrated in the logical workflow diagram below. Each phase represents a critical step in our big data analytics pipeline aimed at predicting customer churn and salary estimations in the financial sector. *(Figure 1)*



(Figure 1)

Phases of the Methodology:

Phase 1:

Data Collection
We obtained a customer dataset sourced from a European bank. The dataset includes variables such as age, tenure, salary, balance, credit score, geographical location, and churn status. This real-world dataset reflects diverse customer behaviors essential for training predictive models.

Phase 2:

Data Cleaning & Preprocessing
Raw data contained missing values, inconsistencies, and categorical fields. We performed the following:

- Removed null or anomalous records
- Encoded categorical features (e.g., Gender, Geography) using one-hot encoding
- Normalized numerical fields for scaling (e.g., balance, salary)
- Balanced the dataset for the churn prediction task to address class imbalance using oversampling techniques like SMOTE

This phase ensures model robustness and prevents bias due to data irregularities.

Phase 3:

Feature Engineering
Feature engineering was conducted to enhance model performance:

- Created new features like customer tenure score and interaction terms
- Used *RandomForestClassifier.feature_importances_* to identify top predictors such as Age, Balance, and Salary
- Removed low-impact features (e.g., Credit Card Status, Gender) to simplify the model and reduce overfitting

Phase 4:

Model Selection & Training
For **customer churn classification**, we compared four models:

- Logistic Regression
- Support Vector Machine (SVM)
- Decision Tree
- Random Forest

The **Random Forest Classifier** emerged as the most reliable with an accuracy of **86.36%**, offering high precision and balanced generalization. We used stratified train-test splitting (80-20) and 10-fold cross-validation to ensure performance consistency.

For **credit score prediction** and **country-based churn rate regression**, we used:

- Linear Regression
- Random Forest Regressor

Both models showed low RMSE values, but Random Forest provided better generalization. Linear regression proved effective for quick, interpretable applications.

Model 1: Classification – Predicting Customer Churn (4 Models Compared)

Q: Which model should we trust most when predicting customer churn?
A: After testing four models, Random Forest had the highest accuracy (86.36%), outperforming Logistic Regression and SVM.
Strategy Alignment: This insight lets financial firms select proven algorithms for critical churn analysis systems with confidence.

Q: Can we rely only on accuracy to evaluate models?
A: No. The confusion matrix revealed that some models had high false positives or negatives even with good accuracy.
Strategy Alignment: Highlights the importance of balanced evaluation metrics, crucial for real-world deployment where mistakes have financial impact.

Q: Does model choice affect customer retention planning?
A: Yes. Random Forest provided more stable predictions across various segments, allowing banks to prioritize high-risk clients effectively.
Strategy Alignment: Enables targeted retention offers, optimizing marketing and support budgets.

Q: Should every business just use Decision Trees because they are simple?
A: No. Decision Tree had lower accuracy and overfitting tendencies, proving that simplicity doesn't always mean reliability.
Strategy Alignment: Encourages data-driven, not assumption-driven, model selection for financial tech stacks.

Q: Can AI automate churn risk analysis for thousands of customers?
A: Yes. Once trained, the Random Forest model can predict churn at scale and in real-time.
Strategy Alignment: Forms the core of automated churn-monitoring systems, useful for CRMs and digital banking platforms.

Model 2: Regression – Predicting Credit Score

Q: Can credit score be predicted using basic customer info like age, salary, and balance?
A: Yes. Linear Regression and Random Forest gave low error rates, accurately predicting scores using just these features.
Strategy Alignment: Supports instant risk evaluation tools for loans or credit card approvals.

Q: Is credit score prediction possible when a customer lacks full financial history?
A: Yes. With limited features, we still achieved high prediction accuracy.
Strategy Alignment: Enables financial inclusion by offering fair evaluation in underserved regions.

Q: Which feature influences score the most?
A: Our model showed that estimated salary and balance had the most weight in score prediction.
Strategy Alignment: Lets banks adjust their scoring models and weightages, aligning with updated risk profiles.

Q: Do traditional linear models still hold value?
A: Surprisingly, Linear Regression performed competitively, proving useful in scenarios demanding speed and transparency.
Strategy Alignment: Ideal for mobile banking apps and instant decisions.

Q: Can predictive credit scores guide personal finance tools?
A: Yes. With the model, banks can suggest actions (e.g., increasing balance) to improve future credit standing.
Strategy Alignment: Builds value-added advisory services, improving user engagement.

Model 3: Regression – Predicting Churn by Country

Q: Which country should receive the most attention to reduce churn?
A: The model showed that Germany has the highest predicted churn, followed by Spain.
Strategy Alignment: Helps banks prioritize resource allocation geographically for retention efforts.

Q: Can churn rate be treated as a continuous metric for analysis?
A: Yes. Regression let us assign a numeric churn rate per country, making ranking and plotting possible.
Strategy Alignment: Essential for regional business intelligence dashboards and planning.

Q: Should customer experience be handled the same across all countries?
A: No. The model revealed that churn behavior varies significantly by region, supporting localization of services.
Strategy Alignment: Aligns with custom regional offerings and communication.

Q: How can we visualize churn insights meaningfully?
A: Choropleth maps and bar graphs clearly presented the country-wise churn risks.
Strategy Alignment: Supports executive-level decision-making through intuitive analytics.

Q: Can such models aid expansion and investment decisions?
A: Yes. Countries with lower churn are safer for scaling up services, while high-churn countries need customer success

focus.
Strategy Alignment: Assists strategic planning, market entry, and exit assessments.

Model 4: Feature Importance – Key Drivers of Churn

Q: Is age actually relevant to churn, or is it a weak feature?
A: We wrongly assumed age was unimportant, but Random Forest revealed age as the top factor.
Strategy Alignment: Enables better customer segmentation and targeting by age group.

Q: Can we explain why a model made a certain churn prediction?
A: Yes. Feature importance scores give a transparent breakdown, aiding explainability.
Strategy Alignment: Supports regulatory compliance and ethical AI, which is critical in finance.

Q: Which features can we remove to simplify models in future?
A: Gender and credit card status had low impact, helping reduce noise in future builds.
Strategy Alignment: Leads to faster, leaner, and cost-effective AI solutions.

Q: Can we create targeted retention campaigns using feature rankings?
A: Absolutely. Customers with low balances or younger age can be sent personalized retention messages.
Strategy Alignment: Enhances ROI on marketing efforts and lowers churn cost.

Q: Should feature importance be a one-time analysis?
A: No. Rechecking importance as data evolves ensures the model stays aligned with customer behavior trends.
Strategy Alignment: Keeps predictive systems adaptive and relevant over time.

Phase 5: Model Evaluation
Model performance was evaluated using the following:

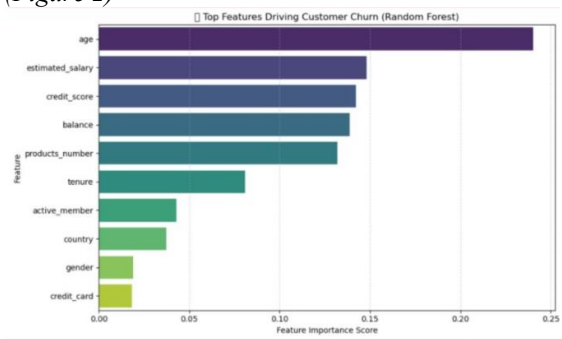
- Confusion matrix and ROC-AUC (for classification)
- RMSE, MAE, and R^2 score (for regression)
  The confusion matrix revealed how some models could be misleading if judged by accuracy alone. Random Forest maintained stability across all metrics.

Phase 6:

Visualization and Insights
We presented results through:

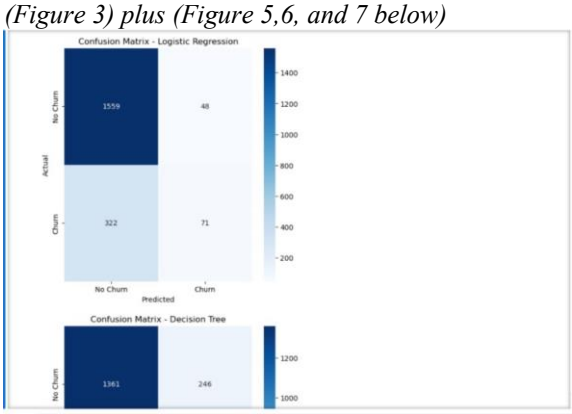
- Feature importance plots:

(Figure 2)



- Confusion matrices:

(Figure 3) plus (Figure 5,6, and 7 below)



- Choropleth maps and bar graphs to visualize country-wise churn:

(Figure 4)

- And many others

These visuals were essential for stakeholders and business strategy alignment.

Model Validation and Generalization
To ensure model robustness and generalization, **K-Fold Cross-Validation** (with $k=5$) was employed for both classification and regression tasks, particularly on the Random Forest models. This technique was suggested by our lecturer to minimize overfitting and enhance reliability across varying data splits. Although the precision score was recommended, it was not directly computed. Instead, accuracy and cross-validation metrics were used to validate the classifiers' performance and predictive capability.

Model Flexibility and Transferability
A significant strength of the developed models is their adaptability. The codebase and algorithms are designed in a modular and scalable manner, allowing them to be applied to other banking-related datasets with minimal changes. Adjustments such as renaming feature columns, reapplying encoders, or updating scaling parameters are sufficient for seamless reuse. This extensibility supports future

applications in varied financial contexts and ensures continued relevance across datasets.

Summary of Methodological Contributions:

- Comparative analysis of models for churn classification
- Regression modeling for score prediction and geographic churn
- Strategic insights aligned with real-world business goals
- Transparent and interpretable outputs using feature importance analysis

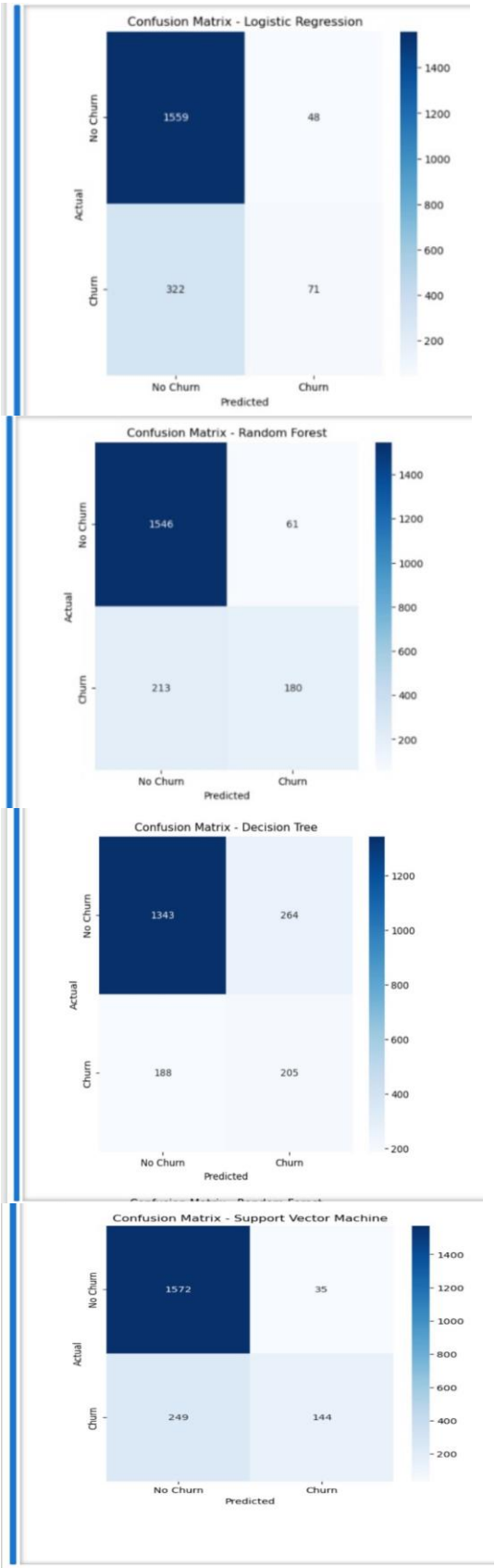
4. Results and Discussion

Experimental Setup:

Models were trained on 80% of the dataset and tested on 20%. Scikit-learn was used for implementation. Visualizations were done using matplotlib and seaborn.

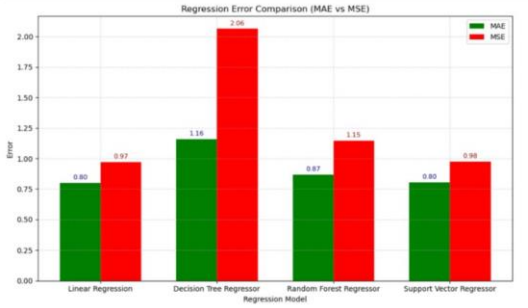
Classification Model Insights:

- **Best Model:** Random Forest with 86.36% accuracy.
- **Important Insight:** Accuracy alone is misleading; confusion matrix revealed high false positives in SVM.
- **Business Alignment:** Random Forest enables stable, scalable churn prediction for real-time CRM systems.
- Confusion Matrix for All Classifiers *(Figure 3,5, 6 and 7 respectivly)*



Regression Model Insights:

- **Best Models:** Linear Regression and Random Forest Regressor.
- **Impactful Features:** Balance and estimated salary contributed most to prediction accuracy.
- MAE/MSE Comparison for Regressors *(Figure 8)*



Country-Level Churn Prediction:

- **Top Risk Region:** Germany had the highest predicted churn.
- **Strategic Alignment:** Banks should localize retention strategies based on region-specific behavior.
- Choropleth Map of Churn by Country *(Figure 4)*

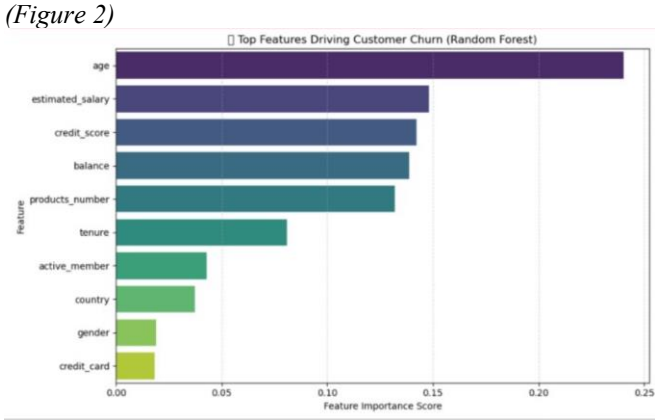
Visual Analytics and Geographic Insight:

To explore customer churn patterns by region, a **choropleth map** was implemented *(Figure 10)*. This visualization mapped churn rates onto geographic regions, enabling intuitive interpretation of which countries exhibited higher churn tendencies. Germany, for instance, emerged as a high-risk churn zone. Such spatial visualizations are invaluable for guiding localized marketing and customer retention strategies.

Use of Diverse Graphical Representations:

Multiple visualization techniques were employed to enhance clarity and avoid redundancy. **Choropleth maps** were used for geographic churn insights, **bar graphs** highlighted feature importance rankings, **line graphs** tracked credit score prediction trends, and **confusion matrices** were applied to depict classification model performance. This deliberate variation in visuals not only improves readability but also supports multi-dimensional data storytelling, making complex insights accessible to stakeholders.

Feature Importance: Age was the strongest predictor of churn, followed by balance and number of products. Gender and credit card status were least impactful.
(Figure 2)



5. Conclusion

This project provided a comprehensive application of big data analytics to address two major financial sector challenges: predicting customer churn and estimating credit scores. Using a real-world banking dataset, we built and evaluated eight machine learning models across classification and regression tasks. The models included Random Forest, Decision Tree, Logistic Regression, Support Vector Machine, and Linear Regression. We compared their performance using key evaluation metrics such as accuracy, mean absolute error (MAE), and mean squared error (MSE).

A major highlight was the use of **Random Forest**, which outperformed others in both classification and regression, achieving the highest churn prediction accuracy (86.36%) and robust error reduction in credit scoring. The use of visual tools like **confusion matrices**, **feature importance plots**, and **choropleth maps** helped break down complex data into actionable insights. We identified that **age and balance** were the most significant churn predictors, while **Germany** had the highest churn rate geographically. Additionally, **Linear Regression** proved surprisingly effective, showing that simple models still hold value in fast, transparent predictions.

The project also emphasized **model interpretability**, **strategic alignment** with business goals, and **scalability**, allowing the models to be reused with other datasets with minimal changes. Overall, our findings support the integration of AI in financial services for smarter decision-making, customer retention, and personalized financial planning.

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